

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2

PAPER 2 November 2022

2 hours 30 minutes

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2022 4004/2.

Additional materials: Mathematical tables or a non-programmable calculator may be used.

INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer ALL questions

1. A sewing machine is marked at \$500 in Bindura. <<
>>(a) 15% discount. Discounted price. [3]<<
>>(b) Then add VAT 15%. Amount paid. [3]<<
>>(c) Deposit 40% of the final amount. Balance remaining. [3] **[9]**

2. 21 Form 4 pupils at Moleli scored:<<
>>2 (3 pupils), 6 (3 pupils), 5 (5 pupils), 6 (6 pupils), 2 (4 pupils).<<
>>(a) Modal mark. [1]<<
>>(b) Mean mark. [4]<<
>>(c) Range. [2]<<
>>(d) One extra pupil scores 6. Effect on the mean? [2] **[9]**

3. (a) Make t the subject of $v = u + at$. [3]<<
>>(b) Simplify $(3x/4) + (x/6)$. [3]<<
>>(c) Solve $2/x = 5/(15)$. [4] **[10]**

4. (a) Using ruler and compasses only, construct triangle ABC with $AB = 7$ cm, $BC = 5$ cm and $AC = 7$ cm. [4]<<
>>(b) Construct the perpendicular bisector of BC. [2]<<
>>(c) Mark the circumcentre and describe its property. [2] **[8]**

5. (a) Expand and simplify $(6x - 2)(x + 2)$. [3]<<
>>(b) Factorise $x^2 + 8x + 15$. [2]<<
>>(c) Solve $(x + 2)/6 = 3$. [3]<<
>>(d) Simplify $(x^2 - 9)/(x - 3)$. [2] **[10]**

6. A cylindrical tank at Moleli has radius 7 cm and height 12 cm. $\pi = 22/7$.<<
>>(a) Volume. [3]<<
>>(b) Curved surface area. [3]<<
>>(c) Total surface area including the base only (open top). [2] **[8]**

SECTION B (48 marks) — Answer any FOUR of the seven questions

- 7.** O is the centre. A, B, C on the circumference. AB is a diameter. Angle BAC = 32° .
 (a) Angle ACB. Reason. [3]
 (b) Angle BOC (centre, same arc BC). [3]
 (c) D on the remaining circumference. Angle BDC. Reason. [3]
 (d) State the angle-in-a-semicircle theorem. [3] **[12]**
- 8.** Triangle P(1,1), Q(3,1), R(1,3).
 (a) Reflect in the y-axis. Coordinates of P'. [4]
 (b) Translate PQR by (3 ; -1). Give Q''. [3]
 (c) Matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ maps (x;y). Describe fully and find the image of Q. [5] **[12]**
- 9.** 60 pupils at Moleli sat a test (0–50). An ogive is drawn.
 (a) How to read the median from the ogive. [3]
 (b) Median estimated as 30. Meaning? [2]
 (c) How to find the IQR from the graph. [4]
 (d) Why IQR is better than the range here. [3] **[12]**
- 10.** (a) y varies directly as x. When $x = 4$, $y = 36$. Find y when $x = 10$. [4]
 (b) z varies inversely as x. When $x = 3$, $z = 8$. Find z when $x = 6$. [4]
 (c) A farmer near Bindura has at most 30 hours. Maize takes 2 h/ha, vegetables 1 h/ha. Inequality for hectares m and v. [4] **[12]**
- 11.** (a) Solve $x^2 - 11x + 24 = 0$ by factorisation. [4]
 (b) A rectangular plot in Bindura is 8 m longer than it is wide. If the area is 33 m^2 , form and solve an equation for the width w. [5]
 (c) Sketch $y = (x - 3)(x - 8)$, showing intercepts. [3] **[12]**
- 12.** A kombi leaves Bindura at 08:00 and travels 120 km at constant speed, arriving after 2 hours. It stops 30 minutes, then returns at 80 km/h.
 (a) Outward speed. [3]
 (b) Time for the return. [3]
 (c) Sketch a distance–time graph, labelling axes. [4]
 (d) Average speed for the whole trip including the stop. [2] **[12]**
- 13.** A msasa tree is observed from A on level ground in Bindura. Angle of elevation of the top is 30° . Distance A to the foot is 40 m.
 (a) Height of the object. Use $\tan 30^\circ = 1/\sqrt{3}$. [4]
 (b) From B, 40 m due East of A, the bearing of the foot is 300° . Sketch. [3]
 (c) A bird sits halfway up. Angle of elevation from A. [5] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).